

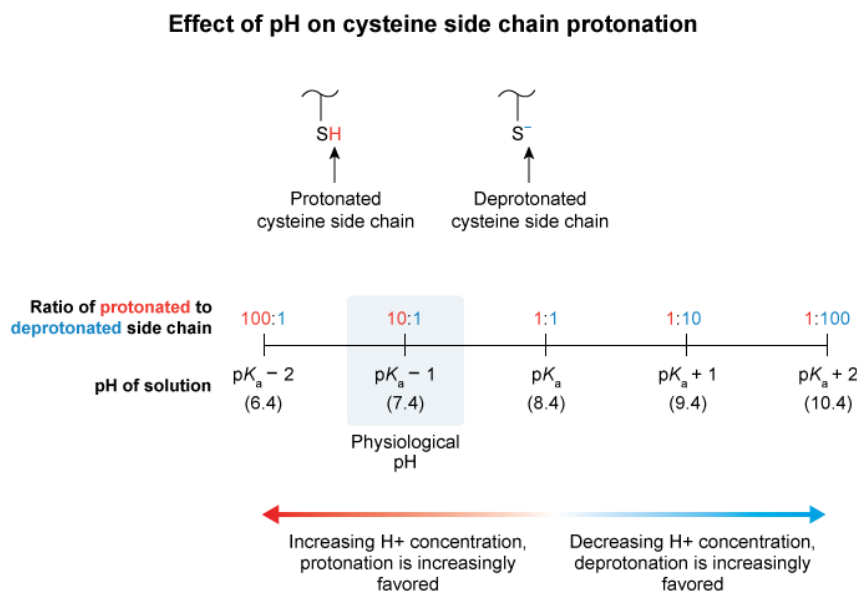
The pK_a for the side chain of cysteine is 8.4. What is the ratio of protonated side chain ($-SH$) to deprotonated side chain ($-S^-$) in a solution of cysteine at physiological pH of 7.4?

- ☐ A. 1:100 [3%]
- ☐ B. 1:10 [18%]
- ✓ ☒ C. 10:1 [66%]
- ☐ D. 100:1 [11%]

Correct

65% answered correctly

Explanation:



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Some chemical groups, including the side chains of some amino acids, can be protonated or deprotonated depending on the pH of the surrounding environment. The **pK_a of a chemical group** is the **pH** at which the **ratio** of protonated

to deprotonated species is **1:1**. A **decrease in pH** yields increased H^+ concentration (**favors protonation**) whereas an **increase in pH** yields decreased H^+ concentration (**favors deprotonation**).

The protonated:deprotonated ratio of a substance can be determined by [solving the Henderson-Hasselbalch equation](#). However, **when the pH differs from the pK_a by an integer number of pH units**, the ratio can be determined more easily because each pH unit **alters the ratio by a factor of 10**.

The question states that the pK_a of the cysteine side chain is 8.4 and the solution is at a pH of 7.4. Because the pH of the solution is lower than the pK_a of cysteine, the protonated form of cysteine will be favored. Furthermore, because the pH is exactly 1 pH unit lower than the pK_a , protonated cysteine will exceed the deprotonated form by a factor of 10. Therefore, the **ratio of protonated to deprotonated cysteine** side chain at physiological pH is **10:1**.

(Choice A) 1:100 is the ratio of protonated cysteine side chain to deprotonated side chain at a pH of 10.4.

(Choice B) 1:10 is the ratio of protonated cysteine side chain to deprotonated side chain at a pH of 9.4. It is also the *deprotonated:protonated* ratio at physiological pH (*reciprocal* of the correct ratio).

(Choice D) 100:1 is the ratio of protonated cysteine side chain to deprotonated side chain at a pH of 6.4.

Educational objective:

The pK_a of a chemical group is the pH at which the ratio of protonated to deprotonated species is 1:1. Decreasing the pH below the pK_a yields a higher proportion of protonated species, and increasing the pH yields a lower proportion. Each pH unit above or below the pK_a alters the ratio of protonated to deprotonated species by a factor of 10.

Time spent:0 sec

Subject:

Biochemistry

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System:

Amino Acids and Proteins